ENReT v1.0.1

Statistical and numerical details to be disclosed, and validation against reference datasets

Technical companion to the user guide: R functions, noteworthy statistical choices and validation

1. Purpose and structure of this document

This document accompanies the user guide of ENReT and makes transparent the aspects of the application that a user, a reviewer or a reader of results obtained with it may need to know. It has three aims:

- to list the R functions, and the packages they belong to, that carry out the statistical computations (section 2);
- to declare the noteworthy statistical and numerical choices (section 3): the choices made by the authors without direct support in the literature, which are stated as such, and the standard techniques applied in a specific way, each with the literature on which it is based;
- to report the validation of the numerical results against certified and published reference data (section 4). Validation is a prerequisite for the use of the application: its results can be trusted only to the extent that they agree with independent reference values.

This document describes version 1.0.1 of ENReT. The validation of section 4 was carried out on version 1.0.0; the least-squares results of version 1.0.1 are identical (verified on all the NIST and Bates & Watts runs), so it applies to this version. The changes of version 1.0.1 are described in items B, I, L, M, N and O and listed in CHANGELOG.md.

2. R functions and packages used for the computations

The table lists the functions that carry out the statistical computations, grouped by package, as verified in the source code of version 1.0.1. All belong to base R (R Core Team; packages stats and base), except nls.lm(), from the minpack.lm package (Elzhov et al.). Elementary functions (sums, means, standard deviations, medians) are not listed. The other packages used by the application perform no statistical computation: shiny and bslib provide the interface, readxl reads Excel files, writexl writes the Excel exports and zip creates the ZIP archives.

Function	Package	Used for
`nls.lm()`	minpack.lm	Least-squares fits with Levenberg-Marquardt: main fit, Monte Carlo refits, Batch, profiles (item F) and secondary models (item G)
`optim()`	stats	BFGS, L-BFGS-B and Nelder-Mead fits (main fit, Monte Carlo refits, Batch); Rigorous Profile with losses other than MSE
`deriv()`	stats	Analytical Jacobian of the model (item I)
`uniroot()`	stats	Exact location of the profile confidence limits (item A); inverse prediction and shelf-life time (item O)
`qf()`, `pf()`	stats	F thresholds of profile intervals and joint confidence regions (item J); p-value of the lack-of-fit test (item V)
`qt()`, `qnorm()`	stats	Critical values of parametric intervals and bands (t with n − p degrees of freedom; normal only when these are not available)
`pt()`, `pchisq()`	stats	p-values of the Welch test and of Cochran's Q (item R)
`p.adjust()`	stats	Holm, Bonferroni and Benjamini-Hochberg corrections (item R)
`quantile()`	stats	Percentile intervals and Monte Carlo bands (R's default sample quantile, type 7; Hyndman & Fan, 1996)
`rnorm()`	stats	Gaussian noise of the Monte Carlo method (item D)

Function	Package	Used for
`shapiro.test()`	stats	Normality test of the bootstrap distributions (item K) and of the inverse-prediction distribution
`bartlett.test()`	stats	Bartlett test of variance homogeneity among replicates (item V)
`lm()`, `anova()`	stats	Brown-Forsythe test (one-way analysis of variance of the absolute deviations from the group medians) and regression of ln(SD) on ln(mean) (items V and W); automatic initial values of the Arrhenius-type secondary model of Non-Isothermal (regression of ln k on 1/T)
`acf()`	stats	Lag-1 autocorrelation of the residuals (item V)
`cov2cor()`, `cor()`	stats	Parameter correlation matrix (from the covariance matrix, or from the Monte Carlo estimates); Pearson correlation between observed and fitted values (item Q)
`approx()`, `approxfun()`	stats	Threshold time of the non-isothermal prediction, common time grid of its bootstrap curves and interpolation of measured temperature profiles (item P)
`density()`, `qqnorm()`, `lowess()`	stats	Graphics only: density curves of the distributions, Q-Q plot of the residuals and smoothing curve of the variance-structure plot (item V)
`sample()`	base	Resampling of the residuals in the Monte Carlo method (item D)
`solve()`	base	Inversion of the scaled covariance matrix (item H)
`choose()`, `digamma()`	base	Exact distribution of the number of runs in the runs test; correction of ln(SD) for the number of replicates (item V)

Least-squares fits. Least-squares fits use the Levenberg-Marquardt algorithm of the minpack.lm package; derivatives are computed symbolically with R's function deriv(). R's nls() is not used by the application: the QR decomposition of the Jacobian (the method of nls(), described by Bates & Watts, 1988) and nls() itself were used only as independent checks (item H and section 4).

3. Noteworthy statistical and numerical choices

Each choice is described as an item, identified by a letter (A–W), with where it is used in the application, what it does and, when relevant, how it was validated and what should be disclosed when results obtained with it are reported. The items are divided into choices without a direct bibliographic source (section 3.2, items A–E), which are choices of the authors, and standard techniques applied in a specific way (section 3.3, items F–W), each with its sources.

3.1 Summary list of noteworthy statistical points

#	Item	Category	Basis / verification
A	Location of the profile confidence limits	Implementation choice	Threshold: Bates & Watts (1988); verified vs 2000-point reference
B	Loss functions other than MSE (MAE, RMSLE)	Implementation choice	Negative values set to 0 with RMSLE (authors' choice); Nelder-Mead recommended with MAE
C	Auto-retry strategy (Batch Elaboration)	Implementation choice	Authors' heuristic; tested on NIST data
D	Monte Carlo data generation	Standard technique	Davison & Hinkley (1997), sections 6.2, 6.3.3, 7.4
E	Bootstrap comparison of two estimates	Implementation choice	Percentile method (Efron & Tibshirani, 1993)
F	Re-optimization in profiles	Standard algorithm	Levenberg-Marquardt; validated on NIST Bennett5
G	Fit of the secondary models	Standard algorithm	Levenberg-Marquardt; validated on synthetic data
H	Hessian inversion with diagonal scaling	Standard technique	Golub & Van Loan (2013); verified vs QR and NIST
I	Analytical Jacobian and covariance	Standard technique	Bates & Watts (1988); R deriv(); verified vs NIST
J	F thresholds for intervals and joint region	Standard technique	Bates & Watts (1988)
K	Monte Carlo intervals and distribution summary	Standard technique	Efron & Tibshirani (1993)
L	Levenberg-Marquardt control settings	Implementation setting	minpack.lm; validated on NIST datasets
M	BFGS, L-BFGS-B, Nelder-Mead control settings	Implementation setting	Nash (2014); validated on NIST datasets
N	Confidence and prediction bands	Standard technique	Bates & Watts (1988, ch. 2); Davison & Hinkley (1997, §7.3–7.4)
O	Inverse prediction and shelf-life time	Standard technique	Delta method / bootstrap percentiles
P	Non-isothermal prediction	Standard technique	Trapezoidal integration
Q	Goodness-of-fit indices	Standard technique	Burnham & Anderson (1998)
R	Comparisons between samples and models	Standard techniques	Welch; Cochran; Holm; Benjamini-Hochberg; Piepho
S	Distribution Density Transformation	Standard technique	ISO 9276-1 (1998); verified by hand calculation
T	Anchor points	Implementation choice	Authors' option; counted as observations
U	Nested bootstrap of the secondary models	Implementation choice	Two-stage propagation of the primary uncertainty

#	Item	Category	Basis / verification
V	Residual and replicate diagnostics	Standard techniques	van Boekel (2008); Carroll & Ruppert (1988)
W	Weighted least squares	Standard technique	Carroll & Ruppert (1988, sections 2.2, 2.9.1, 3.3.6)

3.2 Choices without a direct bibliographic source

These choices are not taken directly from a published source: they are implementation choices or heuristics of the authors, verified where possible on reference data.

A. Location of the profile confidence limits

Where: Profile Likelihood section (Single Fit).

What it does: the profile of each parameter is first computed on a grid of equally spaced points (100 by default, adjustable) within the search range set by the user through the Min/Max fields, pre-filled at ±50% of the estimate (without crossing zero when the estimate is strictly positive or negative). Starting from the estimate, which is always below the F threshold (item J), the grid is scanned outwards until the first point above the threshold on each side; the exact crossing between that point and the nearest point below the threshold is then located with uniroot(), re-computing the profile where needed. The reported interval is the contiguous one containing the estimate: if other regions below the threshold exist beyond it (non-monotone profiles), they are flagged in the table rather than included. If no point above the threshold is found on one side, the limit is set to the edge of the explored range and flagged.

Validation: on Misra1a, Bennett5 and Rat43 (with and without a fixed parameter) the limits agree to about six significant digits with an independent reference obtained from a 2000-point profile with linear interpolation of the threshold crossing, for grid resolutions from 5 to 1000 points; for a one-parameter linear model they coincide with the exact classical interval.

To be disclosed: the threshold follows Bates & Watts (1988); the search strategy (outward scan from the estimate and root finding) is an implementation choice of the authors, analogous to the approach of the profile-based confint() methods in R.

B. Loss functions other than MSE

Where: Single Fit and Batch Elaboration, with the MAE and RMSLE losses.

RMSLE: the RMSLE loss is defined only for non-negative values: observed and predicted values below zero are treated as 0 in the loss. The interface states this, and a warning reports how many observed values are affected.

MAE: the MAE loss is not differentiable, so BFGS and L-BFGS-B, which use derivatives, can stop far from the minimum (16–87% higher loss in tests) or fail (L-BFGS-B, abnormal line-search code). With these combinations the application shows a note next to the choice of the algorithm and a warning at the start of the fit, and recommends Nelder-Mead.

Huber loss (removed in version 1.0.1): in version 1.0.0 the threshold of the Huber loss, 1.345 · MAD of the residuals, was recomputed from the residuals at every evaluation, and the MAD is centred on the median of the residuals rather than on zero: the minimized function was therefore not the loss of the standard Huber estimator, and the result could depend on the starting point and on the algorithm (e.g. residuals shifted away from zero). The loss was removed; operator configurations saved with it switch to least squares, with a warning.

C. Auto-retry strategy (Batch Elaboration)

Where: Batch Elaboration, optional (off by default).

What it does: when the fit of a sample does not meet the acceptance criteria, the fit is repeated from new initial values, up to a maximum number of attempts (25 by default, from 1 to 100), with the algorithm selected by the user. The new initial values are obtained by multiplying each free initial value by 10^{u}, with u drawn from a uniform distribution on [−1.5, 1.5] (a value equal to zero is replaced by a draw from a uniform distribution on [−0.1, 0.1]); with the optional adaptive range, for positive values the interval of u is narrowed so that the new value does not cross the parameter boundaries. A fit is accepted if it has converged, has $R^2 \geq 0$ and adjusted $R^2 \geq 0$ and, with the MSE loss, finite standard errors; optionally, a fit is also rejected if a standard error exceeds a chosen percentage of its estimate. The first accepted fit is kept; if no attempt is accepted, the attempt with the lowest value of the loss is kept.

To be disclosed: random restarts are a common heuristic for multimodal problems, but the specific scheme (log-uniform perturbation, adaptive range, acceptance criteria) is an implementation choice of the authors. In tests with deliberately poor initial values on NIST data sets, about one accepted fit in four was a local rather than the global minimum: the acceptance criteria check convergence, not global optimality, and well-chosen initial values remain essential.

D. Monte Carlo data generation

Where: Monte Carlo error method (Single Fit, Batch Elaboration and Model Comparison).

What it does: at each of N iterations (100 by default, from 10 to 5000) a new data set is generated as $y^* = \hat{y} + e^*$, where $\hat{y}$ are the fitted values, and the model is refitted with the same loss, algorithm and settings. The noise e^* is either (i) a resample with replacement of the residuals (residual bootstrap) or (ii) Gaussian noise with zero mean and standard deviation equal to a value set by the user or, by default, to the RMSE, $\sqrt{(SSE/(n - p))}$ (parametric bootstrap). For prediction bands, new noise of the same type is added to each bootstrap curve. Monte Carlo results are reported only if at least 10 iterations succeed; otherwise the application states it and falls back to the classic standard errors and bands.

Residuals used in the residual bootstrap: by default (option enabled), the modified residuals of Davison & Hinkley (1997), $r_j = e_j/(1 - h_j)^{1/2}$, where h_j is the leverage of observation j (section 6.2.1, eq. 6.9, p. 259): the raw residuals have variance $\sigma^2(1 - h_j)$ (eq. 6.8) and are therefore smaller than the true errors. The modified residuals are centred, $r_j - \bar{r}$, before resampling (section 6.2.3, Algorithm 6.1, pp. 261–262), and the same centred residuals provide the noise of new observations for the prediction band (section 6.3.3, Algorithm 6.4, p. 284). For nonlinear models the leverages come from the linear approximation of the model, with the derivatives of the model at the estimates in place of the explanatory variables (section 7.4, p. 355): h_j are the diagonal elements of $J(J^TJ)^{-1}J^T$, with the analytical Jacobian J of item I. With losses other than MSE only the centring is applied, since the leverage correction is derived for least squares. When the option is disabled, the raw residuals are resampled without centring.

Validation: on a straight-line model, where the linear approximation is exact, the leverages agree with R's hatvalues() to $3 \cdot 10^{-16}$. With 2000 iterations, the bootstrap standard errors with raw residuals were 89–91% of the classic ones on NIST Misra1a (14 points) and on the treated Puromycin data (12 points), and 96–100% with modified residuals, in line with the agreement between residual bootstrap and least-squares theory described by Davison & Hinkley (section 6.2.3). On the BOD data (6 points, strongly nonlinear) the bootstrap standard errors with modified residuals exceed the classic ones by about 20%, a case where the classic, linearization-based standard errors are themselves unreliable.

Sources for regression: bootstrapping the residuals of a regression model, with the covariates kept fixed and new responses $\hat{y} + e^*$, is described by Efron & Tibshirani (1993, chapter 9, section 9.4, pp. 111–112, equations 9.24–9.26), who show that for the linear model it reproduces the usual standard errors, and by Davison & Hinkley (1997, section 6.2.3, Algorithm 6.1). Both books compare it with bootstrapping the pairs (x, y) (Efron & Tibshirani, 1993, section 9.5, pp. 113–115; Davison & Hinkley, 1997, section 6.2.4, Algorithm 6.2, pp. 264–266), which does not assume that the error distribution is the same at every x but changes the design from one bootstrap sample to the next; keeping the covariates fixed is preferable when they are chosen by the experimenter (Efron & Tibshirani, 1993, section 9.7), as is typical of kinetic data. With weighted least squares the residuals are resampled as in item W.

To be disclosed: both schemes are standard (Efron & Tibshirani, 1993; Davison & Hinkley, 1997). Implementation details chosen by the authors: leverages are capped at 0.99 to avoid divisions by values close to zero; Davison & Hinkley also recommend correcting the residuals for the bias due to nonlinearity (section 7.4) without describing the correction, and it is not applied; with strong intrinsic curvature the linear approximation of the leverages may be inaccurate (same section). The minimum of 10 successful iterations is a choice of the authors.

E. Bootstrap comparison of two estimates

Where: pairwise comparisons in Batch Elaboration, Model Comparison and Shelf-Life (two-sample tests and post-hoc comparisons), when both fits have Monte Carlo distributions with at least 10 values.

What it does: the distribution of the difference D between the two parameters is formed from the pairwise differences between the two bootstrap distributions (all $n_1 \times n_2$ differences up to one million; above that, a systematic sample of one million taken at regular steps of the full sequence, which is deterministic and statistically equivalent because the bootstrap iterations are independent and in random order). The confidence interval of the difference is given by the $\alpha/2$ and $1 - \alpha/2$ percentiles of D, and the two-sided p-value by $2 \cdot \min[P(D \leq 0), P(D \geq 0)]$, capped at 1; its resolution is the inverse of the number of differences, and it is 0 (reported as < 0.001) when all differences have the same sign. When at least one fit has no Monte Carlo distribution, the Welch test of item R is used.

To be disclosed: the percentile interval is a standard bootstrap method (Efron & Tibshirani, 1993); forming the difference distribution from all pairs (or a systematic sample of them) and the p-value formula are implementation choices of the authors. On test data the systematic sample gave intervals identical to the full computation up to the third decimal place.

3.3 Standard techniques applied in a specific way

These items use established techniques, with the literature on which they are based; they are described because the specific way in which they are applied affects the results or has to be known to reproduce them.

F. Re-optimization in profiles

Where: Profile Likelihood section (marginal confidence intervals) and Rigorous Profile of the Loss Surface Explorer.

What it does: at each point of a profile the parameter(s) under study are fixed and the remaining free parameters are re-estimated by least squares with Levenberg-Marquardt (nls.lm, settings of item L, analytical Jacobian of item I), starting from their optimal estimates; the profile value is the resulting residual sum of squares. For a model with a single parameter no re-optimization is needed. The procedure is independent of the algorithm chosen for the main fit, and parameter boundaries are not applied. For the Rigorous Profile with a loss other than MSE (not a least-squares problem), BFGS with a Nelder-Mead refinement is used instead.

Validation: on NIST Bennett5 the profile of b1 (100 points) matches, at every point, a reference profile computed independently (nested one-dimensional optimization); the 95% confidence interval, [−3199.0, −2026.5], is identical to the reference. On Rat43 the profile values agree to 10^{-13} with those of an alternative procedure (BFGS with Nelder-Mead refinement).

G. Fit of the secondary models (Shelf-Life and Non-Isothermal)

Where: kinetic model of Shelf-Life (Arrhenius, Q10, Eyring or custom) and secondary model of Non-Isothermal, in Batch Elaboration and Model Comparison.

What it does: least-squares fit with Levenberg-Marquardt (nls.lm, settings of item L), for the point estimate and for every bootstrap refit, independent of the loss and algorithm chosen in the sidebar. The analytical Jacobian (item I) is used when the model is given as a formula (Shelf-Life); for the built-in secondary models of Non-Isothermal nls.lm computes derivatives numerically. Covariance as in item I. The automatic initial values of the Arrhenius-type secondary model come from a linear regression of ln k on 1/T.

Validation: no NIST reference exists for these models. On six synthetic cases (Arrhenius, Eyring, Q10; three or four temperatures; shelf life in days and in minutes), starting from the default initial values, the parameters agree with a reference fit with tight tolerances to 8.6–10.1 digits; one case requires initial values of the right order of magnitude. The four built-in secondary models of Non-Isothermal converge from the automatic initial values on synthetic k(T) data.

H. Hessian inversion with diagonal scaling

Where: covariance matrix and standard errors in Single Fit, Batch Elaboration and the secondary models, for every algorithm.

What it does: the matrix H (J^TJ) is scaled to unit diagonal, $H_s = D^{-1}HD^{-1}$ with $D = \mathrm{diag}(\sqrt{h_{ii}})$, and inverted as $H^{-1} = D^{-1}H_s^{-1}D^{-1}$ with solve(): mathematically identical to direct inversion, but numerically stable. If the diagonal is not positive or not finite, direct inversion is attempted.

Rationale: with parameters on very different scales direct inversion is rejected as numerically singular (for NIST Bennett5 the reciprocal condition number is $1.1\cdot10^{-17}$, below R's threshold of about $2.2\cdot10^{-16}$).

Basis and validation: diagonal equilibration (Golub & Van Loan, 2013). On Bennett5 the result agrees to 8 significant digits with the QR decomposition of the Jacobian (Bates & Watts, 1988); on Rat43 it is identical to direct inversion (relative difference $6\cdot10^{-14}$).

I. Analytical Jacobian and covariance matrix

What it does: the Jacobian of the model is computed symbolically with deriv() and used (i) by Levenberg-Marquardt at every iteration and (ii) for the least-squares covariance matrix, $s^2(J^TJ)^{-1}$ with $s^2 = SSE/(n - p)$, for every algorithm. When the formula contains functions that deriv() does not support (e.g. ifelse, abs, min, max) or the derivatives are not finite, central finite differences are used, with a step equal to the cube root of

machine precision times $|\theta_j|$ (times 1 for a parameter equal to zero, |value| < 10^{-8}, the usual typical-size convention of numerical optimization; with a one-sided difference when the model cannot be evaluated on one side, e.g. at a negative value of a parameter on the bound zero). Since version 1.0.1 the same analytical derivatives are also used for the bands, the parameter sensitivity and the delta-method intervals (items N and O).

Validation: on the NIST formulas tested the analytical and numerical Jacobians agree to 10^{-9}–10^{-11} relative to each column.

To be disclosed: the least-squares covariance formula is standard (Bates & Watts, 1988). An invalid covariance matrix (not computable, or with non-positive variances) is never returned: in that case no covariance, standard errors, confidence bands or delta-method intervals are shown.

J. F thresholds for marginal intervals and joint confidence region

What it does: marginal intervals: $SSE_{min}\cdot[1 + F(1-\alpha; 1, n-p)/(n-p)]$; joint region of two parameters (Rigorous Profile): $SSE_{min}\cdot[1 + 2\cdot F(1-\alpha; 2, n-p)/(n-p)]$, with qf().

Bibliographic basis: Bates & Watts (1988).

K. Monte Carlo intervals and distribution summary

What it does: two distinct intervals are reported: **CI Parametric** = estimate ± t(1−α/2; n−p)·SE, with SE from the covariance matrix (items H and I; not available with losses other than MSE), and **CI Empirical** = α/2 and 1 − α/2 percentiles (quantile(), type 7) of the bootstrap distribution. The two are computed independently. For each parameter the application also reports the mean, median and standard deviation (bootstrap standard error) of the distribution, the bias (mean − estimate) and |bias|/SE, flagged when |bias| ≥ 0.25 bootstrap SE (rule of thumb of Efron & Tibshirani, 1993, chapter 10, Estimates of bias), the Shapiro-Wilk test (Shapiro & Wilk, 1965), and the moment estimators of skewness, $g_1 = m_3/m_2^{3/2}$, and excess kurtosis, $g_2 = m_4/m_2^2 - 3$, not corrected for bias (Joanes & Gill, 1998).

L. Levenberg-Marquardt control settings

What it does: nls.lm is called with the analytical Jacobian and: maxiter = the user's Max Iterations, capped at 1024 (a limit of minpack.lm, stated in the interface); maxfev = max(100000, 10 × maxiter × (p + 1)), so that it is never the effective stopping limit; ftol = ptol = 10^x, with the user-settable exponent x (default −12, instead of the library default of about $1.5\cdot10^{-8}$). Profiles (item F) and secondary models (item G) always use 1024 iterations and 10^{-12}. With a loss other than MSE the application uses Nelder-Mead instead of Levenberg-Marquardt, which minimizes sums of squares only, and states it.

Active bounds: nls.lm keeps the parameters within their bounds by moving them back onto the bound; with the analytical Jacobian this projection is not reflected in the derivatives, so when a parameter ends on a bound the other parameters can stop at a non-optimal point while convergence is reported (Puromycin data, Michaelis-Menten model with Vm ≤ 200: K = 0.05608 instead of the constrained optimum 0.05280). Since version 1.0.1, when a parameter ends on a bound the fit is resumed from that point with the numerical Jacobian, which handles the bound correctly, and the solution with the lower sum of squares is kept; the covariance matrix is still computed with the analytical Jacobian. Without active bounds the results are unchanged (all the NIST and Bates & Watts runs are identical). The refinement is used in the main fit, in the reweighting cycles, in the Monte Carlo refits and in Batch Elaboration; profiles and secondary models use no bounds.

Parameters on a bound: the uncertainty of a parameter on a bound is one-sided, so its standard error and confidence interval, computed as for a free parameter, are not reliable (in the example above, the interval of Vm extends beyond 200). Since version 1.0.1 the application reports these parameters with a warning, marks them "(on bound)" in the parameter table and in an On_Bound column of the exports and of Batch Elaboration, and suggests fixing them at the bound value, removing the bound or using a simpler model.

User-settable tolerance: default and range depend on the algorithm: −12 (from −15 to −4) for Levenberg-Marquardt, BFGS and Nelder-Mead; −9 (from −11 to −4) for L-BFGS-B. The value used is reported in the results

summary and in the exported report. NIST validation with Levenberg-Marquardt (52 runs, at least 4 correct digits): 20 runs at 10^{-4}, 32 at 10^{-6}, 43 at 10^{-8}, 50 at 10^{-10} and below. Looser tolerances never caused a fit to be reported as not converged, so their loss of accuracy is not visible to the user; the interface warns about this.

M. BFGS, L-BFGS-B and Nelder-Mead control settings

What it does: optim() is called with parscale equal to the absolute value of the starting point (1 if zero) and with the user-settable relative tolerance of item L (reltol; for L-BFGS-B, which ignores reltol, the tolerance is converted into its factr parameter).

Bounds: BFGS and Nelder-Mead do not use parameter bounds (only Levenberg-Marquardt and L-BFGS-B do). Since version 1.0.1 a warning is shown when an estimate lies outside the bounds that were ignored; no warning is shown when the estimates are within them, because many registry models have default bounds.

Basis and rationale: parameter scaling with optim is discussed by Nash (2014). It was introduced because without it the optimizers stopped before the optimum (NIST Misra1a, measured at that time: BFGS 1.8 correct digits, L-BFGS-B 1.3). With the current settings (default Max Iterations and tolerance, scaling, and the Nelder-Mead budget and restarts described below), on Misra1a from start 1 the parameters have 5.7 correct digits with BFGS, 5.9 with L-BFGS-B and 5.6 with Nelder-Mead. A much tighter factr for L-BFGS-B was tested and discarded: about half of the fits ended with an abnormal line-search code, without any gain in accuracy.

Nelder-Mead evaluations: in optim(), maxit counts function evaluations for Nelder-Mead (one or two per step of the simplex), not iterations as for Levenberg-Marquardt and BFGS, so the same Max Iterations gave Nelder-Mead a much smaller budget. With the default (200) and the default tolerance 10^{-12}, Nelder-Mead did not converge in 34 of 85 registry models (with Levenberg-Marquardt converging), and the fit was rejected unless non-converged results were accepted. The application therefore uses 10 × Max Iterations for Nelder-Mead (2000 with the default), stated next to the field: in the same test 79 of 85 models converged, with the same accuracy relative to Levenberg-Marquardt (median 6.7 correct digits); 10,000 evaluations gave no further gain (79/85), and a looser tolerance (10^{-8}) converged more often but stopped further from the minimum (26 models with fewer than 4 correct digits, against 5). Fits that converged within the previous budget are unchanged.

Nelder-Mead restarts: the simplex can stall before the minimum and still report convergence when its moves become small, especially on ill-conditioned problems. With the option Restart Nelder-Mead (section 5, shown when Nelder-Mead is selected and on by default) the search is restarted from the solution with a new simplex, scaled on the new point, until the loss no longer decreases (relative change below 10^{-10}), at most 10 times; convergence is that of the last run and the evaluations of all runs are added. The restarts are used in every Nelder-Mead fit: main fit, reweighting cycles, Monte Carlo refits and Batch Elaboration. On the 52 NIST fits (26 datasets, two starting points, default settings, non-converged fits rejected), the accepted fits with at least 4 correct digits rose from 26 to 41 and the accepted but inaccurate ones fell from 14 to 2 (from the distant start 1, 9 to 1; from the closer start 2, 5 to 1); on 86 registry models the converged fits rose from 80 to 84 and the median agreement with Levenberg-Marquardt from 6.6 to 6.9 digits. Without restarts the Monte Carlo refits, which start from the original solution, can stop early and underestimate the variability: on the Chloride data with the Huber loss (available in version 1.0.0) the bootstrap standard errors were 6–15% smaller than with restarts. The cost is about 2.3 times the time of a single fit (about 0.1 s per fit on the registry models) and about 1.7 times with Monte Carlo error estimation. Restarts were also tested for BFGS and L-BFGS-B and discarded: they increased the number of accepted but inaccurate fits, because the limiting factor of these methods is the numerical gradient.

NIST validation of the general optimizers: on the 52 NIST fits with default settings (non-converged fits rejected), BFGS accepted 40 fits, of which 21 with at least 4 correct digits, L-BFGS-B 45 and 23, and Nelder-Mead with restarts 43 and 41; Levenberg-Marquardt reaches at least 4 correct digits in all fits except those of Lanczos1, whose certified residual sum of squares is at the limit of double precision (section 4.1). On difficult problems the general optimizers can stop far from the minimum while reporting convergence, so Levenberg-Marquardt remains the recommended and default algorithm for least squares; the general optimizers are needed for losses other than MSE, where Nelder-Mead is selected automatically.

N. Confidence and prediction bands

What it does: linear method: bands by the delta method, $\hat{y}(x) \pm c \cdot SE(x)$, with $SE(x)^2 = g^T \cdot Cov \cdot g$ for the confidence band and $SE(x)^2 + s^2$ for the prediction band, where g is the gradient of the model with respect to the parameters, computed analytically (the derivatives of the fit; numerically only for formulas that cannot be differentiated symbolically, item I). By default the bands are pointwise, with $c = t(1-\alpha/2; n-p)$. Optionally (off by default) the confidence band can be simultaneous: the 1−α approximate inference band of Bates & Watts (1988, chapter 2), $f(x, \hat{\theta}) \pm s \cdot \|v^T \hat{R}^{-1}\| \cdot \sqrt{(p \cdot F(p, n-p; \alpha))}$, where v is the vector of derivatives at x and $\hat{R}$ the triangular factor of the QR decomposition of the derivative matrix; since $s \cdot \|v^T \hat{R}^{-1}\| = \sqrt{(v^T \cdot s^2 (V^T V)^{-1} \cdot v)} = SE(x)$, only the multiplier changes, $c = \sqrt{(p \cdot F(p, n-p; \alpha))}$, with $F(p, n-p; \alpha)$ the upper α quantile, $qf(1-\alpha, p, n-p)$. The prediction band is always pointwise. Monte Carlo method: α/2 and 1 − α/2 percentiles of the bootstrap curves (confidence band) and of the bootstrap curves with added noise (prediction band, item D), always pointwise. These are the pointwise percentile-method confidence limits of a fitted curve described by Davison & Hinkley (1997, section 7.3, pp. 352–353, and, for functions of a nonlinear regression curve computed from the simulated parameter estimates, section 7.4, pp. 356–358), who use R = 999 bootstrap replicates; the application uses 100 by default, which can be increased up to 5000.

Secondary models: the plots of the secondary models (kinetic model of Shelf-Life, secondary model of Non-Isothermal) show the bands of the nested bootstrap (item U) when it is available and, otherwise, linear bands computed in the same way from the covariance matrix of the secondary model (item G), with the same optional simultaneous band; the prediction band adds the residual variance of the secondary model. The lower limit of the Non-Isothermal band is truncated at zero.

Validation: on NIST Misra1a the half-width of the simultaneous band computed by the application agrees with the formula of Bates & Watts evaluated independently through the QR decomposition and analytical derivatives to $5 \cdot 10^{-7}$ (value measured in version 1.0.0, when the gradient was computed numerically). For the secondary models, on an Arrhenius kinetic model of shelf life (seven temperatures) and an Arrhenius secondary model of the rate constant (six temperatures), the pointwise and simultaneous half-widths agree with a hand calculation of the delta method to 10^{-16} and $2 \cdot 10^{-9}$ respectively.

To be disclosed: pointwise bands hold at each value of x separately; the simultaneous band holds for the whole curve at once and is wider (with n = 6 and p = 2, multiplier 3.73 instead of 2.78). For nonlinear models both are approximations based on the linearization of the model (Bates & Watts, 1988).

Version 1.0.1: the bands previously used central finite differences with a step proportional to the parameter: for a parameter equal to zero (e.g. a lag on its bound) the step was about 10^{-14} and the bands became very wide and irregular (Reparameterized Richards on ComBase B092_25: width of the confidence band 1.5–16.5 with jumps of up to 235% between neighbouring points, against 0.34–0.78 with analytical derivatives). With the analytical derivatives the other bands differ from version 1.0.0 only by 10^{-12}–10^{-9} in relative terms.

Weighting: with weighted least squares the confidence band uses the weighted covariance matrix and the prediction band the variance $\sigma^2 \cdot \hat{y}(x)\text{^}\zeta$ of a new observation (item W).

O. Inverse prediction and shelf-life time

What it does: the value $\hat{X}_0$ at which the fitted curve reaches a target value (inverse prediction) or a quality threshold (shelf-life time) is found by scanning the search range (200 points) for every crossing of the target and refining each crossing with uniroot(), with a tolerance of 10^{-10} relative to the size of the range; crossings through a vertical asymptote, where the model is not close to the target, are discarded. With more than one solution the smallest X is returned and the user is warned. Inverse prediction requires a valid search range (pre-filled with the X range of the data). Linear method: delta method for inverse regression, $Var(\hat{X}_0) = [g^T \cdot Cov \cdot g + s^2$ (prediction interval only)$] / f'(\hat{X}_0)^2$. Monte Carlo method: $\hat{X}_0$ is recomputed for every bootstrap parameter set, taking the solution closest to the main one, and the interval is given by its percentiles. For a new observation (prediction interval of inverse prediction, "New Observation" in Shelf-Life) a resampled residual is added to the target value at every iteration: the same leverage-corrected and centred residuals as the prediction band (Davison & Hinkley, 1997). The derivatives of the delta method are analytical (item I).

To be disclosed: the delta-method interval is a first-order approximation (Seber & Wild, 1989).

Version 1.0.1: previously only the two ends of the search range were checked (non-monotone curves could give "no solution"), Shelf-Life used the default absolute tolerance of uniroot() (about 10^{-4}), inverse prediction resampled the raw residuals (slightly too narrow intervals) and, in Shelf-Life, the "New Observation" choice applied only to the samples fitted with the linear method.

Weighting: with weighted least squares the variance of a new observation at the target value is $\sigma^2 \cdot y_target\hat{}\zeta$, in both the delta method and the Monte Carlo method (item W).

P. Non-isothermal prediction

What it does: for a time-temperature profile T(t), the cumulative integral of the rate constant, $K(t) = \int k(T(s))ds$, is computed with the trapezoidal rule on a grid of 500 points; the predicted quantity follows from the integrated rate law of the chosen order, and the time at which a threshold is reached is obtained by linear interpolation of K(t) with approx(). Monte Carlo curves are brought to a common time grid by linear interpolation before computing percentiles.

To be disclosed: standard approach for non-isothermal kinetics (van Boekel, 2008); the number of integration points is fixed by the application.

Q. Goodness-of-fit indices

What it does: $R^2 = 1 - SSE/SST$; adjusted $R^2 = 1 - (1 - R^2)(n - 1)/(n - p)$; $RMSE = \sqrt{(SSE/(n - p))}$; $CV(RMSE) = 100 \cdot RMSE/\bar{y}$; Pearson correlation between observed and fitted values; information criteria in the least-squares form of Burnham & Anderson (1998), with $K = p + 1$ estimated parameters (the p model parameters plus the error variance): $AIC = n \cdot \ln(SSE/n) + 2K$; $AICc = AIC + 2K(K + 1)/(n - K - 1)$; $BIC = n \cdot \ln(SSE/n) + K \cdot \ln n$.

To be disclosed: this form omits the constant $n \cdot (1 + \ln 2\pi)$ of the Gaussian log-likelihood, so the absolute values differ from those of R's AIC() and BIC() applied to an nls fit by exactly that constant (verified on NIST Misra1a); differences between models fitted to the same data, Akaike weights and evidence ratios are identical. The information criteria assume maximum-likelihood estimates, i.e. least squares with Gaussian errors: with losses other than MSE they are not computed (in Batch Elaboration the columns are removed, in Model Comparison they are shown as N/A, with a note). R^2 is reported for familiarity, but it is not a reliable measure of fit for nonlinear models (Spiess & Neumeyer, 2010); model comparison should rely on AIC, AICc or BIC (Burnham & Anderson, 1998).

Weighting: with weighted least squares SSE, σ and R^2 are the weighted ones and the information criteria include the term $-\Sigma \ln w_i$ of the weighted likelihood (item W).

R. Comparisons between samples and models

Where: two-sample tests and post-hoc comparisons of Batch Elaboration, Model Comparison and Shelf-Life.

Two estimates without Monte Carlo: Welch test: $t = |\hat{\theta}_1 - \hat{\theta}_2| \,/\, \sqrt{(SE_1^2 + SE_2^2)}$, with Welch-Satterthwaite degrees of freedom (Welch, 1947; Satterthwaite, 1946), in which each sample contributes $n - p$ degrees of freedom; two-sided p-value with pt(). With Monte Carlo distributions, item E.

Omnibus test: Cochran's Q (Cochran, 1954): $Q = \Sigma\, w_i\, (\hat{\theta}_i - \bar{\theta}_w)^2$, with weights $w_i = 1/SE_i^2$ and inverse-variance weighted mean $\bar{\theta}_w$, compared with a χ^2 distribution with $k - 1$ degrees of freedom (pchisq()). The standard errors are the bootstrap ones when all compared fits have Monte Carlo distributions, the linear ones otherwise; in Batch Elaboration and Shelf-Life each sample uses the best uncertainty available (bootstrap where present). A Monte Carlo distribution with fewer than 10 valid values is not used, either in the omnibus test or in the pairwise comparisons (as in item D).

Multiple comparisons: the pairwise p-values are adjusted with p.adjust() separately for each parameter, with the method chosen by the user: Holm (1979; default), Bonferroni, or Benjamini & Hochberg (1995), which controls the false discovery rate instead of the family-wise error rate. Verified against p.adjust() on the post-hoc of Model Comparison.

Compact Letter Display: two samples share a letter when they are not significantly different; the letters correspond to the maximal cliques of the graph of non-significant pairs, enumerated with the Bron-Kerbosch

algorithm (Bron & Kerbosch, 1973), which handles non-transitive cases (a sample compatible with two others that differ from each other); this follows the principle of the letter-based display of Piepho (2004).

S. Distribution Density Transformation

Where: data input of Single Fit and Batch Elaboration (optional, off by default), for sieve or particle-size data.

What it does: the y values are treated as the amount retained on each sieve, whose material lies between that sieve and the next larger one, so that each sieve is a boundary of its class and not its centre; the largest class (oversize) needs an assumed upper boundary or is discarded, and the smallest point keeps its own value as lower boundary unless the user assumes or discards it. The density of each class is its amount divided by the class width, on a linear scale (Δx) or a logarithmic scale ($\Delta \ln x$, natural logarithm), and by the total amount of the original data, including any discarded class; the representative x of a class is the arithmetic or geometric mean of its boundaries. Classes of invalid width (e.g. a boundary equal to zero on the logarithmic scale) are excluded.

Basis and validation: the convention follows ISO 9276-1 (1998) for the representation of particle-size distributions. On an example with six classes (pan and sieves from 63 to 1000 µm) the representative values and densities agree with a hand calculation; on the linear scale the area under the density is 1, and on the logarithmic scale the pan class is excluded while its amount remains in the total, so that the area of the kept classes equals the retained fraction (0.95 in the example).

T. Anchor points

Where: Distribution Density Transformation (optional, off by default).

What it does: one or two artificial points with density 0, at x = 0 and/or at a chosen x, are added to the data after any transformation, to constrain the fitted curve towards zero beyond the measured classes.

To be disclosed: the anchor points are not measurements, but they are treated as observations in every subsequent computation: number of points, degrees of freedom, standard errors and confidence intervals, goodness-of-fit indices, information criteria and Monte Carlo resampling. The reported uncertainty is therefore smaller, and R^2 larger, than the measured data alone would justify; the interface states this. The option is an implementation choice of the authors.

U. Nested bootstrap of the secondary models

Where: Shelf-Life (kinetic model) and Non-Isothermal (secondary model), in Batch Elaboration and Model Comparison, when the bootstrap of the secondary model is selected.

What it does: at each iteration one value per temperature (shelf life or rate constant) is drawn from the uncertainty of the corresponding primary fit: from its Monte Carlo distribution if available, otherwise from a normal distribution with mean equal to the estimate and standard deviation equal to its linear standard error (a point estimate without uncertainty is kept fixed). The secondary model is refitted to the drawn values with Levenberg-Marquardt (item G), and the distribution of its parameters (and, for Non-Isothermal, of the predicted critical time) gives percentile intervals. At least 10 successful iterations are required.

To be disclosed: this two-stage scheme propagates the uncertainty of the primary estimates into the secondary model; it is an implementation choice of the authors. Two limitations: (i) the scatter of the values around the secondary model beyond the primary uncertainty (lack of fit) is not resampled, whereas the linear intervals of the secondary model include it through the residual variance, so the two methods measure partly different sources of uncertainty; (ii) draws from the normal approximation can be negative, which is physically impossible for a rate constant or a shelf life: their frequency is negligible when the standard error is below about 30% of the estimate (< 0.05%), but reaches 2%, 9% and 16% when it is 50%, 75% and 100% of the estimate.

V. Residual and replicate diagnostics

Where: Single Fit, below the main plot: Residual Analysis, normal Q-Q plot, lag plot and Replicate SD vs Mean plot (2 × 2 grid), followed by the replicate diagnostics.

Replicates: observations are grouped by x (values compared after rounding to 12 significant digits, so that equal values differing only by representation errors are not separated). With n observations, f distinct x values and p parameters, the pure-error sum of squares, which measures the model-independent experimental error, is $SS_{pe} = \Sigma_i \Sigma_j (y_{ij} - \bar{y}_i)^2$, with n − f degrees of freedom.

Lack-of-fit test: as described by van Boekel (2008, chapter 7, section 7.4.3): the residual sum of squares SSr is split into pure error and lack of fit, $SS_{lof} = SSr - SS_{pe}$ with f − p degrees of freedom, which measures the inability of the model to fit the data; the ratio of the lack-of-fit and pure-error mean squares, $F = [SS_{lof}/(f - p)] / [SS_{pe}/(n - f)]$, is compared with F(f − p, n − f), and the sums of squares are summarized in an ANOVA table. A large ratio suggests that the model does not fit the data adequately. For nonlinear models the test is approximate, as are the other F-based procedures (Bates & Watts, 1988). Computed only with the MSE loss, when n > f and f > p.

Variance homogeneity: Bartlett's test (R's bartlett.test()) over the x values with at least two replicates, which assumes normally distributed data, and the Brown-Forsythe test, i.e. a one-way analysis of variance of the absolute deviations from the median of each group, over the x values with at least four replicates. With two replicates the median equals the mean of the pair and the two absolute deviations are identical, so the test is undefined (on the Puromycin data, with duplicates, it returned p = 0); with three, the median coincides with one of the values, whose deviation is zero, and in simulations the test never rejected. In simulations with eight groups and equal variances the false-positive rate of Bartlett's test was 4.3–4.8% and that of the Brown-Forsythe test 7.0% with four replicates per x, against a nominal 5%.

SD vs mean: for the x values with replicates and positive mean and standard deviation (at least three), the exponent of the power-of-the-mean variance model, $SD \propto mean^{\theta}$, is estimated by regressing ln(SD) on ln(mean) (the method of Rodbard described by Carroll & Ruppert, 1988, section 3.3.6, p. 89); the application calls the slope b and reports its 95% confidence interval, stating which of the reference values b = 0 (constant variance), 0.5 (variance proportional to the mean) and 1 (constant coefficient of variation) it includes. Following the same section, ln(SD) is corrected for the number of replicates m, since for normal data $E[\ln s] = \ln \sigma + c(m)$: the application subtracts $c(m) = \tfrac{1}{2}[\psi((m - 1)/2) - \ln((m - 1)/2)]$, with ψ the digamma function, which gives −0.635, −0.289 and −0.184 for m = 2, 3 and 4 (the values reported in the book). With the same number of replicates in all groups the correction changes only the intercept; with different numbers it removes a bias of the slope (in 500 simulations with constant variance and two or six replicates per group, the mean b was −0.317 without correction and −0.028 with it). The plotted standard deviations are the corrected ones. The plot and the replicate diagnostics also report ζ = 2b with its interval, the exponent of the variance model $var(y) = \sigma^2 \cdot \hat{y}^{\zeta}$ used by the weighting (item W).

Variance structure without replicates: when fewer than three x values have replicates, the fourth plot shows instead $|r|^{(2/3)}$, the cube root of the squared studentized residuals $r = e/(s\sqrt{(1 - h)})$, against the logarithm of the predicted values, with a lowess smoothing curve (Carroll & Ruppert, 1988, section 2.7, pp. 29–31, and Figures 2.1–2.3): with normal data and constant variance $|r|^{(2/3)}$ is the Wilson-Hilferty transformation of a chi-squared variable, and a rising trend indicates variance increasing with the mean. The leverages h come from the Jacobian of the model; on a straight line, where the linear approximation is exact, the studentized residuals agree with R's rstandard() to $3 \cdot 10^{-15}$.

Lag plot: each residual against the previous one, ordered by x; with replicates the mean residual per x is used, since the order of replicates at the same x is unknown. The lag-1 autocorrelation is computed with R's acf() and compared with the approximate 95% bounds $\pm 1.96/\sqrt{m}$ (NIST/SEMATECH e-Handbook of Statistical Methods).

Runs test: the runs test on the signs of the residuals described by van Boekel (2008, section 7.8, Goodness of Fit and Model Discrimination), on the residuals ordered by x (mean residual per x with replicates; zero residuals excluded). With n_p positive and n_n negative residuals and n_R observed runs, the expected number of runs $R_r = 2n_p \cdot n_n/(n_p + n_n) + 1$ and its variance $\sigma_r^2 = 2n_p \cdot n_n(2n_p \cdot n_n - n_p - n_n)/[(n_p + n_n)^2(n_p + n_n - 1)]$; the book prints the last factor of the denominator as $(n_p - n_n - 1)$, which gives zero or negative variances and was taken to be a typographical error (the formula used agrees with simulated variances, e.g. 2.73 against 2.76 for six positive and six negative residuals, where the printed formula gives −30). The standard normal deviates $Z_{tf} = (n_R - R_r + 0.5)/\sigma_r$ and $Z_{tm} = (n_R - R_r - 0.5)/\sigma_r$

are reported with their sign (strongly negative Z_tf: too few runs; strongly positive Z_tm: too many). Van Boekel suggests about three standard deviations as a signal; since |Z| cannot reach 3 with fewer than about 14 residuals, even with only two runs, the interpretation is based on the exact one-sided p-values of the distribution of the number of runs (Swed & Eisenhart), which agree with 100,000 random permutations. When even the most extreme pattern could not give $p < 0.05$ (e.g. six residuals, minimum $p = 0.10$), the interface says so. In 400 simulations with 20 points the test was significant in 1.8% of fits of the correct model (conservative because the distribution is discrete) and in 95.0% of fits of a straight line to curved data; on the Chloride data it gives 16 runs against 27.7 expected (Z_tf = −3.11, $p = 0.0008$).

Batch Elaboration: the same diagnostics are available for each sample of a batch, below the Goodness-of-Fit Diagnostics table, after pressing Compute residual diagnostics: a summary table with one row per sample (lag-1 autocorrelation, runs test, lack-of-fit test, Bartlett and Brown-Forsythe tests, b and $\zeta = 2b$), the 2 × 2 grid and the tables of a selected sample, or the plots of all samples arranged by sample or by plot type, with single, combined and ZIP downloads. Each sample is converted to the form of a Single Fit result, with the residuals recomputed from the formula exactly as in the batch fit, and processed by the same functions; on simulated data with four samples, with and without weighting, the diagnostics of every sample agreed exactly with those of the Single Fit on the same data. The summary table notes that, with many samples and several tests per sample, some significant results are expected by chance alone. The diagnostics are cleared when the batch is run again.

Runs test in Model Comparison: the comparison table reports, for each model, Z_tf and the exact p-value for too few runs, computed in the same way (residuals ordered by x, mean residual per x with replicates), so that models fitted to the same data are treated identically. It is meant as a check of systematic patterns, not as a criterion for choosing the model: models with more parameters tend to leave more random residuals, so among the models without systematic patterns the information criteria should be used. On the Chloride data the straight line gives 5 runs against 28.0 expected ($p \approx 10^{-11}$), and even the three-parameter model of the book, the best by AICc, gives too few runs ($p = 0.0008$), reflecting the serial correlation of these data discussed by Bates & Watts (1988).

Validation: on the treated Puromycin data (duplicates at six concentrations) the lack-of-fit F (1.0709, $p = 0.4468$) and Bartlett's statistic agree exactly with independent R computations (pure error from lm(y ~ factor(x)), residual sum of squares from nls(), bartlett.test()), and the lag-1 autocorrelation agrees with acf(); on the Chloride data it is 0.67, outside the bounds, consistent with the serial correlation described by Bates & Watts (1988). In 500 simulations with a correct model and constant variance, the lack-of-fit test and Bartlett's test were significant in 4.2% and 5.0% of cases (nominal 5%), and the mean estimated b was 0.00; with standard deviation proportional to the mean, b averaged 0.99.

To be disclosed: according to Carroll & Ruppert (1988, section 3.3.6 and Table 3.6, p. 90), the regression of ln(SD) on ln(mean) has a low efficiency with few replicates (20% with two, 41% with three, 53% with four relative to pseudo-likelihood), so b is imprecise with two or three replicates per x, and regressing on sample rather than true means is an errors-in-variables problem, usually minor. The lack-of-fit test assumes constant variance: in the simulations with standard deviation proportional to the mean it was significant in 9.6% of cases even with the correct model. With few replicates per x the variance homogeneity tests have low power, so a non-significant result does not prove that the variance is constant.

W. Weighted least squares

Where: Single Fit and Batch Elaboration, selector Weighting (least squares) in section 5 of the sidebar, available only with the MSE loss: none; $w = 1/\hat{y}$ ($\zeta = 1$, variance proportional to the mean); $w = 1/\hat{y}^2$ ($\zeta = 2$, constant coefficient of variation); $w = 1/\hat{y}^{\zeta}$ with ζ set by the user; $w = 1/\hat{y}^{\zeta}$ with $\zeta = 2b$ estimated from the replicates.

Variance model: $\mathrm{var}(y_i) = \sigma^2 \cdot f(x_i, \theta)^{\zeta}$, the power-of-the-mean model of Carroll & Ruppert (1988), with weights $w_i = 1/\hat{y}_i^{\zeta}$, where $\hat{y}_i$ is the value predicted by the model; all predicted values at the data points must be positive.

Estimation: generalized least squares with full iteration, i.e. iteratively reweighted least squares (Carroll & Ruppert, 1988, section 2.2, pp. 13–14): starting from the unweighted fit, the weights are computed from the predicted values, the model is refitted by weighted least squares with fixed weights, and the two steps are repeated until the largest relative change of the parameters is below 10^{-9} (at most 30 cycles; a warning is shown if the weights do not stabilize). In the weighted fit each residual and the corresponding row of the Jacobian are multiplied by $\sqrt{w_i}$, so Levenberg-Marquardt is used unchanged; BFGS, L-BFGS-B and Nelder-Mead minimize $\Sigma\, w_i(y_i - \hat{y}_i)^2$. The final weights, computed from the final fit, are kept fixed for all subsequent inference.

ζ from the replicates: ζ = 2b, where b is the slope of the regression of ln(SD) on ln(mean) of the groups of replicates, with the correction for the number of replicates (item V; Carroll & Ruppert, 1988, section 3.3.6, p. 89), computed with the same code as the Replicate SD vs Mean plot. In Batch Elaboration ζ can be common to all samples (the groups of replicates of all samples in a single regression, which assumes the same variance structure in all samples, as when they are measured with the same analytical method; implementation choice) or estimated for each sample from its own replicates.

Standard errors and indices: covariance matrix $s^2(J^TWJ)^{-1}$ with the weighted Jacobian, for every algorithm, with $s^2 = SSE_w/(n - p)$ and $SSE_w = \Sigma\, w_i(y_i - \hat{y}_i)^2$; σ = s is the scale of the variance model (for ζ = 2, the relative standard deviation). R^2 is computed with the weighted sums of squares about the weighted mean (implementation choice); CV(RMSE) is not reported. The information criteria use the Gaussian log-likelihood with known relative variances, which contains the additional term $\tfrac{1}{2}\Sigma \ln w_i$, so AIC = n·ln(SSE_w/n) − Σ ln w_i + 2K (K = p + 1), comparable between fits with different variance structures.

Bands and prediction: the confidence band uses the weighted covariance matrix (also the simultaneous band of item N). The prediction band uses the variance of a new observation, σ²·ŷ(x)^ζ plus the variance of the fitted curve, as in Carroll & Ruppert (1988, section 2.9.1, equations 2.37–2.38, pp. 52–53), who note that ignoring the heterogeneity makes the prediction interval too long where the variability is small and too short where it is large. Inverse prediction, which section 2.9 treats as calibration, and the delta-method shelf-life time use σ²·y_target^ζ, and so do the models saved in Model Comparison, which keep their weights.

Monte Carlo: the residual bootstrap resamples the weighted residuals $\sqrt{w_i}(y_i - \hat{y}_i)$, with the leverage correction of item D computed from the weighted Jacobian, and rescales each drawn residual to its point by dividing by $\sqrt{w_i}$; the Gaussian method uses standard deviation $\sigma/\sqrt{w_i}$. The bootstrap refits use the same fixed weights. This is Algorithm 6.3 of Davison & Hinkley (1997, section 6.2.6, Non-constant variance: weighted error resampling, pp. 270–272), with V(μ̂) = ŷ^ζ: modified residuals e/{V(μ̂)(1 − h)}^½, with the leverages of the weighted fit, new responses μ̂ + V(μ̂)^½·δ*, and weighted least squares with the empirical weights 1/V(μ̂) from the fitted values; Efron & Tibshirani (1993, section 9.7, model 9.42, pp. 120–121) apply the same idea, resampling residuals scaled by the factor that describes how the error grows with the dose.

Profiles and grid: the Profile Likelihood and the Grid Search use the weighted sum of squares with the fixed weights of the fit, with the same F thresholds.

Diagnostics: the residual plot, the normal Q-Q plot, the lag plot, the runs test and the variance-structure plot without replicates use the weighted residuals, which should have constant spread if the weights are right; the lack-of-fit test uses the weighted residual sum of squares and the weighted pure error (the weight is constant within each x value). The variance homogeneity tests and the Replicate SD vs Mean plot remain on the observed data, since they serve to choose the weights.

Validation: without weighting the results are bit-for-bit identical to those before the change in all fields (parameters, standard errors, covariance, criteria, Monte Carlo distributions, bands, diagnostics, Grid Search functions), in eight Single Fit configurations and four Batch configurations. With ζ = 2 on simulated data with 5% relative error, the estimates and standard errors agree with an independent iteratively reweighted fit with R's nls(weights =) to 7.5–8.6 digits, and AIC(nls) − AIC differs by exactly n(1 + ln 2π); the weighted profile agrees with nls() to 10^{-16}, and the weighted inverse prediction and delta-method shelf-life time agree with hand calculations. Each sample of a weighted Batch gives exactly the Single Fit result, and the common and per-sample ζ agree with regressions computed by hand. In 250 simulated data sets with 8% relative error (ζ = 2) and three replicates per x, the 95% intervals of the rate constant covered the true value in 94.8%

(unweighted), 96.0% (ζ = 2) and 94.0% (ζ = 2b) of cases, and the weighted estimates were more than three times more precise (standard deviation 0.0018 against 0.0059); ζ = 2b averaged 2.01.

To be disclosed: as in standard theory, the weights are treated as fixed (Davison & Hinkley, 1997, section 6.2.6). The uncertainty of an estimated ζ is not propagated to the standard errors (in the simulations above the mean standard error was 0.00182 against a standard deviation of the estimates of 0.00199); with two to four replicates per x, ζ estimated from the replicates is imprecise, which is why Batch Elaboration offers a ζ common to all samples; the common ζ assumes the same variance structure in all samples; weighting requires positive predicted values at all data points and is not applied with losses other than MSE.

4. Validation

Validation is a prerequisite for the use of the application. The numerical results were compared with certified values (NIST, section 4.1), with exact least-squares solutions computed independently (Bates & Watts, section 4.2, and Carroll & Ruppert, section 4.3) and with the results of a reference software for predictive microbiology (ComBase, section 4.4). The complete results are given in the accompanying workbook ENReT_v1.0.0_validation_results.xlsx.

4.1 NIST Statistical Reference Datasets

The 26 NIST StRD nonlinear regression datasets with a single predictor were used (Nelson, with two predictors, is not applicable); the data included in the application were compared value by value with the original NIST files and are identical. The validation harness executes the server function of the deployed application file with simulated Shiny reactivity and calls the same computation triggered by the "Run Optimization Calculation" button, with each NIST model entered as a custom formula and the starting values as the initial-guess string a user would type. Settings: Levenberg-Marquardt, MSE loss, linear error method, Start 1 and Start 2, Max Iterations = 1024; R 4.3.3, minpack.lm 1.2.4.

Accuracy is expressed as the log relative error, LRE = $-\log_{10}(|\text{estimate} - \text{certified}|/|\text{certified}|)$, the number of correct significant digits (McCullough, 1998, 1999), for the parameters, their standard errors and the residual sum of squares (minimum over the parameters). LRE ≥ 4 for all three is considered acceptable; cells below 4 are shaded.

Dataset	Difficulty	Start 1 (LRE)			Start 2 (LRE)		
		Param.	SE	RSS	Param.	SE	RSS
Misra1a	Lower	9.8	9.5	10.5	10.1	7.3	10.5
Chwirut2	Lower	7.5	6.6	11.2	7.2	6.3	11.2
Chwirut1	Lower	8.5	7.4	11.4	7.5	6.2	11.4
Lanczos3	Lower	7.2	7.2	10.6	6.4	6.4	10.6
Gauss1	Lower	8.7	6.7	11.6	8.6	8.4	11.6
Gauss2	Lower	10.0	8.1	10.6	9.9	8.1	10.6
DanWood	Lower	10.0	8.1	11.7	10.7	10.7	11.7
Misra1b	Lower	9.5	9.3	11.3	9.3	9.1	11.3
Kirby2	Average	7.4	6.7	11.6	8.0	7.2	11.6
Hahn1	Average	7.5	7.1	10.6	10.7	10.4	10.6
MGH17	Average	8.1	6.1	11.5	8.1	6.2	11.5
Lanczos1	Average	10.6	3.0	2.7	10.6	3.3	3.0
Lanczos2	Average	10.4	8.7	10.1	9.1	8.9	10.3
Gauss3	Average	10.4	10.3	11.0	9.5	7.5	11.0
Misra1c	Average	10.6	8.0	11.1	10.1	7.3	11.1
Misra1d	Average	8.7	8.5	11.2	12.6	11.1	11.2
Roszman1	Average	8.6	7.5	12.2	8.6	9.1	12.2
ENSO	Average	5.1	6.1	11.4	5.2	6.1	11.4
MGH09	Higher	6.1	6.0	11.6	6.0	5.9	11.6
Thurber	Higher	6.7	5.9	11.2	6.6	5.8	11.3
BoxBOD	Higher	7.3	6.6	10.4	7.1	6.4	10.4
Rat42	Higher	9.2	7.6	11.9	8.3	6.8	11.9
MGH10	Higher	9.8	10.0	11.5	10.9	10.2	11.4

Dataset	Difficulty	Start 1 (LRE)			Start 2 (LRE)		
		Param.	SE	RSS	Param.	SE	RSS
Eckerle4	Higher	9.7	8.4	10.7	9.6	8.3	10.7
Rat43	Higher	7.5	6.6	11.4	7.0	6.3	11.4
Bennett5	Higher	11.0	7.4	11.1	11.0	7.4	11.2

Each cell is the minimum over the parameters (or over their standard errors) of that run, i.e. the LRE of the least accurate parameter; the LRE of every single parameter is given in the NIST sheet of the accompanying workbook ENReT_v1.0.0_validation_results.xlsx.

The validation was repeated after the last additions to the application (weighted least squares, residual diagnostics, Nelder-Mead restarts): with Levenberg-Marquardt all 26 datasets gave, for both starting points, exactly the same numbers as in the table above.

50 of 52 runs reach LRE ≥ 4. **Lanczos1**: the parameters agree to 10.6 digits; the certified residual sum of squares, $1.4 \cdot 10^{-25}$, cannot be reproduced to several relative digits in double precision, and the standard errors, which depend on it, reach about 3 digits.

Other algorithms (same datasets, both starting points, default settings, non-converged fits rejected): accepted fits with LRE ≥ 4 in 21 of 52 runs with BFGS, 23 with L-BFGS-B and 41 with Nelder-Mead with restarts (item M); these general-purpose optimizers may also report convergence short of the certified solution, so Levenberg-Marquardt is the recommended algorithm for least squares. The 26 NIST models of the "Models for Reference Data" tab, selected from the model registry with their Start 1 values, reach 5.1–11 correct digits.

4.2 Bates & Watts (1988) data sets

Fourteen data sets of the book are included as reference data sets (data from the R package NRAIA and R's base datasets, verified value by value); their models and the reference values given in the notes were checked by the authors against the book. There are no certified values for these data, so for the 15 models of the "Models for Reference Data" tab (the three-compartment model of the lipoprotein data is not included) the reference is the exact least-squares minimum computed independently in R with minpack.lm (nls.lm with tolerances of 10^{-16}, restarted from its own solution), with standard errors $s^2(J^TJ)^{-1}$ at the minimum; the estimates agree with the values printed in the book to the digits given there. This reference uses the same MINPACK Levenberg-Marquardt library as the application, with much tighter tolerances, which explains the identical values (LRE = 15) of some models; an independent check is given by R's nls(), a Gauss-Newton algorithm, whose residual sums of squares agree with the application to at least 10.4 digits in all models. The application was used with the models of the tab and their starting values, derived from the data with the graphical methods of the book (Levenberg-Marquardt, Max Iterations = 1024). As in section 4.1, each cell is the minimum LRE over the parameters or over their standard errors.

Model (Models for Reference Data)	Parameters (min LRE)	SE (min LRE)	RSS (LRE)
Puromycin, treated cells	8.5	7.3	15.0
Puromycin, untreated cells	7.2	6.4	13.7
BOD	15.0	7.8	15.0
BOD2	8.6	7.7	15.0
Chloride ion transport	15.0	14.0	15.0
Growth of leaves	6.3	5.5	13.7
Count Rumford's cooling experiment	15.0	7.8	15.0
Pharmacokinetics of sulfisoxazole	7.8	6.8	14.2
Pharmacokinetics of tetracycline	7.7	6.6	14.0

Model (Models for Reference Data)	Parameters (min LRE)	SE (min LRE)	RSS (LRE)
Pharmacokinetics of saccharin	8.0	6.7	14.0
PCB in Lake Cayuga trout	15.0	14.6	15.0
Nitrite utilization, day 1 (model 1)	7.5	6.5	14.2
Nitrite utilization, day 1 (model 2)	7.0	5.4	14.2
Nitrite utilization, day 2 (model 1)	15.0	6.4	15.0
Nitrite utilization, day 2 (model 2)	7.3	5.8	15.0

For the PCB data the book fits the linear model ln(y) = β1 + β2·x^(1/3) on transformed variables (β1 = −2.391, β2 = 2.300), reproduced in the application with the ln(y) pre-transformation; fitting y = exp(β1 + β2·x^(1/3)) on the original scale gives different estimates because it assumes an additive rather than multiplicative error. The reference values given in the notes of the data sets were computed earlier with R's nls(), which stops with a looser convergence criterion; for validation the exact minima of the workbook should be used.

4.3 Weighted least squares: Carroll & Ruppert (1988) esterase assay data

The esterase assay data of Carroll & Ruppert (1988, Table 2.3, pp. 46–47, Example 2.2, pp. 48–51) are included as a reference data set for weighted regression: counts in a binding assay against the concentration of the enzyme esterase, with severe heterogeneity of variance, fitted with the linear model y = β0 + β1·x and the power-of-the-mean variance model with constant coefficient of variation (the book's θ = 1, i.e. ζ = 2 in the application). The 113 observations were transcribed from the page images and compared value by value with the text layer of the PDF, with complete agreement; the 5 observations marked by an asterisk are excluded, as in the book, leaving 108.

With ζ = 2 the application gives β0 = −37.98 (SE 12.23), β1 = 18.180 (SE 0.947), σ = 0.2772, identical to an independent iteratively reweighted fit with R's lm(weights =), and without weighting β0 = −17.41, β1 = 17.066, identical to lm(). The book reports β0 = −37.36 (SE 12.21), β1 = 18.16 (SE 0.95), σ = 0.27 and, without weighting, β0 = −17.02, β1 = 17.05. The differences are small relative to the uncertainty (0.6 against a standard error of 12.2 for β0, 0.02 against 0.95 for β1; the standard errors agree within 0.2%) and are already present in the unweighted fit, which involves no weighting choice: they therefore come from the data or the computation of the authors, not from the weighting. No change of a single value of x or y reproduces both the unweighted and the weighted published estimates, and no fixed number of reweighting cycles reproduces the weighted ones; the most likely explanation is that the data file used by the authors differed slightly from the printed table.

Weighting	Quantity	Reference (R lm)	Application	LRE	Book
None	b0	-17.41219996	-17.41219996	15.0	-17.02
None	b1	17.06621288	17.06621288	15.0	17.05
None	SE(b0)	20.51059409	20.51059409	15.0	–
None	SE(b1)	0.8927878489	0.8927878489	15.0	–
None	σ	101.8467200	101.8467200	15.0	–
$1/\hat{y}^2$ (ζ = 2)	b0	-37.98184269	-37.98184271	9.4	-37.36
$1/\hat{y}^2$ (ζ = 2)	b1	18.18009771	18.18009771	10.2	18.16
$1/\hat{y}^2$ (ζ = 2)	SE(b0)	12.23402099	12.23402099	10.4	12.21
$1/\hat{y}^2$ (ζ = 2)	SE(b1)	0.9471381763	0.9471381763	11.3	0.95
$1/\hat{y}^2$ (ζ = 2)	σ	0.2772015907	0.2772015907	11.5	0.27

Reference: R lm() without weights, and with weights 1/fitted² iterated to convergence. Book: Carroll & Ruppert (1988), Example 2.2.

4.4 Agreement with ComBase: Baranyi & Roberts model

The 26 ComBase growth curves of Salmonella in broth included in the application were compared with the parameters estimated by ComBase itself. ComBase fits the model of Baranyi & Roberts (1994), $y(t) = y0 + \mu max \cdot A(t) - (1/m) \cdot \ln[1 + (\exp(m \cdot \mu max \cdot A(t)) - 1)/\exp(m \cdot (ymax - y0))]$, with $A(t) = t - \lambda + \ln(1 - \exp(-vt) + \exp(-v(t - \lambda)))/v$ and $n = v/\mu max = 1$, $m = 1$, to the natural logarithm of the counts, and reports y0, ymax and µmax converted to log10 (ComBase, 2025, Appendix). The same model written for log10 counts was entered in the application as a custom formula; it is mathematically identical, and fitting the application's Integrated Baranyi Model to the counts converted to natural logarithms gave the same estimates to 7–8 digits. For the two curves fitted by ComBase without an upper asymptote, the model $y = y0 + \mu max \cdot A(t)$ was used.

Series	log N0 (%)	Lag (%)	µmax (%)	log Nmax (%)	Max |z|	RMSE / SE of Fit
B092_25	+0.4	−10.9	−3.1	−0.7	0.82	0.904
B092_45	−1.4	−10.5	−3.8	0.0	0.22	0.994
B092_4	+0.5	−2.6	−2.0	0.0	0.25	0.998
B092_32	+0.1	−15.7	−3.5	−0.4	0.40	1.002
B092_17	+0.8	−3.6	−3.9	0.0	0.44	1.016
B146_1	0.0	−3.9	−6.5	0.0	0.11	0.998
B146_2 *	+3.4	+16.3	+0.9	–	0.06	0.993
B132_1	−0.1	−8.2	−12.0	0.0	0.47	0.995
B132_2 *	+5.9	+14.4	+0.5	–	0.15	0.980
B419_12	+0.4	−13.7	−5.1	−0.2	1.40	0.725
B301_75	+0.8	−5.1	−1.9	−0.1	0.20	0.994
B301_79	+0.6	−4.9	−2.2	0.0	0.60	0.956
B301_82	−0.2	−5.8	−2.6	0.0	0.39	0.984
B301_81	−0.5	−11.2	−3.1	0.0	0.56	0.960
B092_26	−2.9	−18.1	−6.7	+0.1	0.58	0.971
B092_30	−1.6	−15.1	−6.7	+0.3	0.71	0.978
B092_43	+0.4	−2.0	−3.1	0.0	0.20	0.954
B092_14	+0.7	−4.2	−4.1	−0.1	0.42	1.003
B092_56	+2.6	+2.2	−2.5	0.0	0.64	0.940
B092_108	+1.2	−6.5	−5.2	0.0	1.85	0.977
B092_109	+0.7	−7.2	−5.7	0.0	2.09	1.013
B092_27	−0.4	−13.2	−4.6	−0.4	1.46	1.017
B092_16	+0.6	−4.8	−4.6	−0.1	1.08	1.009
B092_10	+0.9	−5.0	−3.1	−0.2	1.12	0.972
B104	+7.4	−11.9	−5.4	−1.6	1.22	0.966
B130	−1.3	−69.0	−3.5	−1.1	1.57	0.700

Differences of the application from ComBase in per cent; Max |z| is the largest difference over the parameters of the curve in ComBase standard errors (shaded above 1); RMSE / SE of Fit compares the residual standard errors (below 1: the application's fit has the smaller residuals). * Curves fitted without an upper asymptote.

All 26 fits converged. The initial and maximum levels agree closely with ComBase (median absolute difference 0.7% and 0.05%). The lag and µmax of the application are systematically a few per cent lower (typically 2–18% and 2–12%), but within the uncertainty reported by ComBase: over the 102 parameters the median absolute difference is 0.22 ComBase standard errors, 74% differ by less than 0.5 and 89% by less than 1 standard error, with a maximum of 2.1. With the same model, the residual standard error of the application ($\sqrt{[SSE/(n - p)]}$,

the quantity reported by ComBase as SE of Fit) is equal to or lower than ComBase's in 20 of 26 curves, so the published ComBase parameters are not the exact least-squares minimum for these data, as examined in detail for series B419_12 below. The comparison is therefore presented as agreement with a reference software, in per cent and in standard errors rather than as LRE. The full results are given in the accompanying workbook ENReT_v1.0.0_validation_results.xlsx, together with those of sections 4.1–4.3.

Detailed checks on series B419_12. The ten data points used by ComBase are identical to those of the application. Fitted with the linear model, which has a closed-form solution and involves no iterative algorithm, the two programs agree in every reported digit (initial value 2.552 ± 0.452, rate 0.0333 ± 0.00349, SE of Fit 0.768), which confirms that the data, the standard errors and the SE of Fit are computed in the same way; the quantity reported by ComBase as R-square is the adjusted R^2 (0.909, against R^2 = 0.919). With the Baranyi model, evaluating the ComBase parameters with the curvature parameters n = m = 1 reproduces the ComBase SE of Fit (0.113 against 0.108, a difference compatible with the rounding of the published parameters), while every other combination of n and m gives 0.14–0.20. At the least-squares minimum of the same model found by the application the residual standard error is 0.078 against 0.108: the sum of squares of the ComBase parameters is 2.08 times larger, and they are worse also by the sum of absolute residuals (0.651 against 0.525) and by the largest residual (0.204 against 0.110), so ComBase does not minimize a different criterion but stops before the minimum. ComBase does not document its nonlinear least-squares algorithm; since any algorithm run to convergence reaches the same minimum, the differences are attributed to the convergence of its fitting procedure (stopping criteria, starting values, automatic handling of the curvature parameters) in the long, flat valley created by the correlation between the lag and μmax. The values are given in the sheet ComBase_B419_12 of the workbook.

Integrated Baranyi Model of the application. The model is written for the natural logarithm of the counts, as recommended by ComBase (2025) and Rockaya & Baranyi (2025). With log10 counts its lag and μmax differ from the standard values (by about 17–21% in median on the ComBase curves), so log10 data should be converted to natural logarithms (multiplying by ln 10 = 2.302585) and the results converted back, or the log10 form described above should be entered as a custom formula.

4.5 Web version (Shinylive)

The numerical results of sections 4.1–4.3, obtained in native R, were confirmed by the author in the deployed WebAssembly (Shinylive) version, in Microsoft Edge and Mozilla Firefox, with all LRE ≥ 5 except the known case of section 4.1 (Lanczos1).

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